

A Systematic Literature Review on A Star Schema-Based Cybersecurity Data Warehouse for Phishing and Social Engineering Knowledge Discovery

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Abstract

Phishing and social engineering attacks still pose substantial cybersecurity threats that exploit human vulnerability rather than system weakness. There is a vast amount of research dedicated to phishing and social engineering detection via machine learning, deep learning, and artificial intelligence tools. However, little attention is paid to structuring and analyzing such attacks' data via data warehousing and analytical processing for knowledge discovery purposes. This systematic literature review will consider how the use of star schema-based data warehousing, Online Analytical Processing (OLAP), and data mining approaches may be applied in the cybersecurity domain. Following the PRISMA 2020 guidelines, 46 papers written from 2021 to 2026 were analyzed. The review focuses on three research questions concerning star schema data warehouses' design, OLAP applications in multidimensional analysis, and cybersecurity threat prediction and classification with data mining approaches. The results show that there is an evident advancement in phishing detection and cyber threat intelligence. Nevertheless, research in the field is fragmented and does not provide much information about data warehousing, analytical processing, and predictive modeling applications. Research gaps were identified and discussed, and a concept for designing a star schema-based cybersecurity data warehouse was proposed.

Keywords: Phishing Detection, Social Engineering, Cybersecurity Data Warehouse, Star Schema, OLAP, Data Mining, Knowledge Discovery.

1 Introduction

Rapid digitization of organizational operations and a massive increase in the volume of online data have changed the cybersecurity landscape, bringing new possibilities and challenges. Modern information systems are vulnerable to social engineering attacks, which exploit human psychology rather than software weaknesses or network infrastructure. Unlike traditional cyberattacks, which exploit software vulnerabilities to access sensitive information and resources, social engineering attacks manipulate people into sharing credentials, transferring money, or performing other actions that violate cybersecurity requirements. Recent empirical data demonstrate the seriousness of the situation. Various studies suggest that phishing, spear-phishing, baiting, pretexting, and other types of social engineering are still among the most common cybersecurity threats as they use people’s psychology, not technical weaknesses [1], [2]. Another extensive systematic review of social engineering attacks confirms that these attacks use psychological manipulations, trust abuse, and other aspects of human psychology to make their victims give away their confidential information or take some other actions [2].

Social engineering techniques have undergone a revolution as far as the use of technology is concerned. In modern attacks, the attackers use Artificial Intelligence (AI) and Machine Learning (ML) for generating context-aware messages through Natural Language Processing (NLP) and behavioral profiling that are capable of defeating spam filters as well as deceiving even security-aware individuals. The attackers use deepfakes technology for pretexting, AI voice technology for phishing campaigns, and automated behavioral profiling for water-holing attacks. Even though social engineering as a major cybersecurity problem is increasingly recognized today, there are still many issues in terms of the research literature. First, existing research studies have focused mainly on technical ways of dealing with social engineering—machine learning-based detection methods, phishing email classification, and domain name validation [3], [4], [5]. On the other hand, most research conducted in this field does not emphasize the systematic analysis and organization of social engineering information to discover new knowledge [6]. Second, even though classifications of social engineering attacks are available, there have been no efforts in terms of creating structures to store and analyze such data.

The value of cybersecurity data in analysis relies largely on how effectively the data is organized and turned into significant analytical messages and actionable knowledge. Studies carried out in Cyber Threat Intelligence (CTI) and knowledge discovery have shown that structured analysis approaches provide substantial assistance in mining useful information from cybersecurity datasets [6], [7]. In light of this, designing a star-schema-based cybersecurity data warehouse seems to be a promising solution for supporting systematic cybersecurity analytics and knowledge discovery. With such an approach, one would be able to store and organize information about phishing and social engineering attacks, apply Online Analytic Processing (OLAP) analysis to detect multidimensional patterns, and even integrate data mining to predict and classify threats. In terms of the modern methods used in empirical research, the star-schema-based approach corresponds to the need for structured data processing and extraction of economically meaningful signals.

The star-schema-based approach can manage phishing and social engineering attacks by storing and organizing data, performing OLAP analysis for multidimensional pattern recognition, and employing data mining for predictive threat classification. Nevertheless, the deployment of such an approach entails several problems that must be overcome. Cybersecurity datasets are usually diverse in nature, huge in volume, and dynamic in nature, requiring highly scalable architectures that facilitate efficient multidimensional analysis and knowledge discovery. The application of other data mining methods, such as clustering, classification, association rules mining, and anomaly detection, allows discovering useful patterns in cybersecurity datasets that can reveal some irregularities that might be caused by cyber-attacks, malware attacks, or insider threats [6], [7], [8].

Therefore, the goal of this systematic literature review is to provide an overview of existing research on phishing and social engineering attacks, and on data warehousing, OLAP-based analysis, and threat prediction and classification using data mining approaches. Using the PRISMA 2020 criteria to ensure the transparency, reproducibility, and unbiased selection of the articles, the study addresses the following questions:

- RQ1: How can a star schema-based data warehouse be designed to efficiently store and organize phishing and social engineering data for cybersecurity analysis?
- RQ2: What OLAP techniques can be applied to the data warehouse to identify meaningful patterns and trends in phishing and social engineering attacks?
- RQ3: Which data mining techniques integrated with the star schema warehouse can best predict and classify phishing and social engineering threats for knowledge discovery?

This systematic review seeks to uncover the reliable design principles, provide guidance on methodological issues, reveal research gaps that require further investigation by systematically reviewing evidence extracted from chosen papers, analyzes the state-of-the-art against three research questions and suggests the possible research direction in the form of a star schema-based cybersecurity data warehouse. The key contribution of this research is the incorporation of social engineering studies and phishing research into the area of data warehousing, OLAP and data mining in the context of cybersecurity. It also points out important areas in which further research is needed, especially the gap that exists between threat detection and structured data processing, and suggests a conceptual framework for cybersecurity data warehousing using the star schema model. This serves as a base for future research related to scalable, analytical, and proactive threat analysis in cybersecurity.

2 Literature Review

Social engineering attacks research has seen substantial development in recent years, shifting from early descriptive analysis of attack types to more sophisticated and empirically driven approaches to the analysis of attack patterns, detection techniques, and mitigation strategies [2], [3]. This progression was enabled by an increased realization of the importance of the human factor as a major weakness in cybersecurity

systems, as well as an increased availability of empirical data collected through both real-life cases and simulation of social engineering attacks. An analogous change happened in the domain of data warehousing and analytics with the emergence of systematic approaches to the organization and analysis of security data. At the same time, studies in the field tend to concentrate on single issues such as algorithms for detecting phishing attempts, factors affecting people’s psychological vulnerability, mitigation strategies, and data warehousing design separately. Specifically, a dedicated study on systematic organization and analysis of phishing and social engineering data for the purposes of knowledge discovery via an integrated approach to data warehousing and analysis is not currently available. This section provides a brief overview of empirical results obtained during previous studies.

2.1 Insights From Research Articles

The empirical literature on phishing, social engineering, cybersecurity analytics, and knowledge discovery shows that there is rapid methodological development owing to innovations in machine learning, deep learning, natural language processing, and cybersecurity data engineering. In all these domains, there is always emphasis on the shift from rule-based detection systems to intelligent systems that are adaptive and capable of dealing with complex cyber threats. These findings may be grouped into five main themes: social engineering and human factors, phishing detection methods, cyber threat intelligence and knowledge discovery; cybersecurity data warehousing and dimensional modeling, and AI-based prediction of cyber threats.

2.1.1 Social Engineering Attack Taxonomies and Classification

One of the primary problems encountered by cybersecurity researchers is the classification of social engineering attacks to facilitate the comparison of data and its systematic organization. Taxonomies that classify attacks according to the mechanism of action, psychology involved, and the technological vectors have been proposed by the researchers [2]. In addition, [9] offers a detailed taxonomy of social engineering tactics and security measures that demonstrate the fact that modern attackers tend to use cognitive biases such as authority, urgency, scarcity, and trust when crafting their attack strategies. Moreover, [10] contributes to the same point of view, explaining that user training should be supplemented by some technological means to become effective. Longtchi et al. [1] further this discussion by examining internet-based social engineering through a psychological and behavioral lens, illustrating how attacks become successful owing to the interplay of cognitive weaknesses of humans and technological possibilities. Li et al. [11] add to this discussion by proposing security pattern-based defense frameworks that help align attack approaches with corresponding mitigation tactics. Further research by Schmitt and Flechais highlights a major development in the area of threats, driven by the emergence of generative AI [12]. The results obtained by the authors clearly show that attackers use large language models, deep-fakes, and voice synthesis to develop highly personalized and contextually relevant phishing content.

2.1.2 Phishing Detection Techniques

Machine Learning Approaches

Still, traditional machine learning algorithms remain robust and effective. In particular, Prasad and Chandra introduced the PhiUSIIL model, which merges the concepts of similarity index analysis and incremental learning to increase the efficiency of detecting phishing URLs in dynamically changing conditions [13]. Additionally, [14] proved that a combination of raw URL attributes and HTML-based attributes increases the effectiveness of phishing webpage detection. Similarly, Anirudh et al. presented an ensemble classification model for detecting phishing emails; the combination of several classifiers was examined to increase the efficiency of the detection process and minimize false positives [15]. Collectively, it can be stated that properly designed feature-based models are still relevant in spite of the rising popularity of deep learning techniques.

The authors in [16] develop a hybrid machine learning approach to detect phishing attacks and prove that the use of ensemble approaches based on multiple classifiers increases detection accuracy by using URL and content-based features. Moreover, [17] demonstrates that intelligent machine learning algorithms trained with structured URL and HTML features can be used to distinguish legitimate from phishing websites effectively with high accuracy. Kapan and Gunal illustrate the sensitivity of classifiers to feature engineering techniques in that classifier performance can be highly improved by proper design of URL, domain, and content features [18]. Aljofey et al. developed an advanced technique that integrates both URL and HTML feature engineering and proved that using structural and lexical features improves the detection process against obfuscated phishing websites [19]. Likewise, [20] presents a hybrid approach for phishing website detection using a variety of URL, domain, and webpage features and found that integration of all types of feature categories outperforms the performance achieved using only one set of features. All these studies support the importance of rich input representation in machine learning-based phishing detection systems.

Divakaran and Oest prove that machine learning and deep learning algorithms outperform rule-based systems due to their complexity; however, they note the limitations of such algorithms, including low interpretability and biased datasets [4]. Their research is mainly focused on the effectiveness of machine learning techniques to detect cyber threats is related not only to the efficiency of the algorithms used, but also to the quality of the data, which is used to train those algorithms. Specifically, authors in [21] investigate the use of NLP in phishing email detection and emphasize that linguistic feature extraction, semantic analysis, and text classification models are essential for detecting email-based phishing. They demonstrate that phishing emails have some distinct linguistic characteristics, which include urgent language, authoritative statements, and peculiar syntax.

Deep Learning Architectures

Zara et al. find that the use of convolutional neural networks (CNNs) is highly effective for phishing website detection, with algorithms automatically learning hierarchical structures from input data [22]. Similarly, [23] demonstrates that neural-network-based models can effectively detect phishing URLs by capturing the complex nonlinear

dependencies of URLs, thus confirming the relevance of applying deep learning algorithms for phishing detection problems. Supporting these findings, [24] suggests a hybrid approach based on the use of deep CNNs and Random Forest and demonstrates that combining deep feature extraction and ensemble classification leads to enhanced phishing websites detection performance. Further research by [25] improves the approach by adding the multi-head self-attention mechanism to CNN. This attention mechanism helps the model to concentrate on the most relevant parts of the input data and provides some interpretability of the results due to attention weights evaluation.

Moreover, Hussain et al. propose CNN-Fusion – a lightweight deep learning framework that allows achieving high detection accuracy while remaining efficient computationally and thus usable in real time [26]. The proposed CNN-Fusion architecture utilizes various convolution streams with varying kernel sizes for capturing patterns from different levels. This paper highlights the aspect of computational efficiency, which is a practical consideration when using a particular model in reality. Barik et al. [27] improve deep learning-based phishing detection by utilizing optimization methods for enhancing the convergence and robustness of the models. Similarly, Alsubaei et al. presented a hybrid deep learning architecture for phishing detection and cyber-crime forensics, illustrating how using different parts of deep learning simultaneously may lead to better detection performance along with forensic abilities [28].

Transformer-Based Approaches

The use of transformers has led to impressive progress in detecting phishing attempts in recent years. Using transformer-based models, Meléndez et al. found that transformer-based models outperformed classical ones in classifying phishing emails [29]. Through an extensive empirical evaluation, it has been established that transformer-based methodologies, due to their capabilities of capturing context in textual data, perform far better than classical methodologies. In addition, [30] proposes transformer-based models enhance transparency and detectiveness. Otieno et al. prove that Bidirectional Encoder Representations from Transformers (BERT) based models are very efficient in detecting phishing URLs as they can understand the semantics of textual structures [31]. Li et al. propose KnowPhish, a multi-modal technique that combines large language models and knowledge graphs to provide an improvement to phishing detection robustness under adversarial attack [32]. Nevertheless, [33] note that phishing attacks continually update their evasion methods, and thus it is essential to create detectors that will be more than just static feature analysis. Hence, several studies state that despite dominating the field at the moment, deep learning and transformer-based models face the same challenges in the form of adversarial robustness, data imbalance, and domain generalization problems [34], [4], [5].

2.1.3 Cyber Threat Intelligence and Knowledge Discovery

According to [6], the CTI process is essential in building a proactive defense against cybersecurity attacks, as it allows translating cybersecurity data into intelligence. In particular, the paper discusses different techniques of CTI mining, such as clustering,

classification, and correlation analysis, and proves that extracting structured intelligence from cybersecurity data enables organizations to be prepared for potential risks. The researchers indicate that the implementation of CTI should include not only analytical tools but also intelligence-sharing frameworks. The work of Rahman et al. [8] provides new insights into CTI by focusing on the mining of temporal attack patterns, which enables predictive modeling of cyber threats based on the identification of recurring attack behavior over time. Thus, one can see that cybersecurity data include temporal regularities that can be exploited. Finally, Molina-Coronado et al. focus on Knowledge Discovery in Databases (KDD) techniques employed in intrusion detection systems and show that a structured analytical pipeline, consisting of pre-processing, transformations, and mining, greatly improves detection results [7]. They also highlight the role of anomaly detection and clustering approaches in finding unknown attack patterns. As a result, all of these studies show that cybersecurity datasets include hidden structures that can be revealed by employing data mining and machine learning techniques.

2.1.4 Data Warehousing and Dimensional Modeling for Cybersecurity

The use of data warehousing technology in cybersecurity analysis stands out as an important methodological innovation that facilitates organizing and analyzing security data in multiple dimensions. [7] illustrates the possibility of performing multidimensional analysis of cybersecurity events using data warehouse architecture and OLAP-based decision making. It becomes evident from the work that data warehouse technology is capable of helping analysts perform multidimensional security data analysis according to various dimensions such as time, source, attack types, and severity. Moreover, data warehouses oriented at cybersecurity include features related to privacy preservation, access control, and auditing that permit the safe handling of cybersecurity data. All the above-mentioned is necessary in order to use data warehouses in operation because cybercrime analysis data is usually highly sensitive. Nevertheless, current research concentrates on generic cybercrime analysis, but not on phishing and social engineering.

2.1.5 Artificial Intelligence and Advanced Threat Prediction

In [35] and [36], the use of massive phishing datasets and threat intelligence databases for conducting phishing analysis has been shown to offer insights about the distribution and evolution of phishing attacks. The use of large-scale data analysis allows for a better understanding of attack patterns, targets, and defense effectiveness, helping companies to direct their resources and efforts to the most important problems. The work carried out by Asiri et al. is one of the examples of the development of real-time phishing detection systems, which shows that AI-based detection systems are capable of working well in this environment [37]. In this particular case, the authors prove the operational requirement for real-time security, which means that low latency is required from such systems to provide high levels of detection accuracy.

This domain of research has been further developed by Jayakeerthi et al., who introduce the idea of a hybrid CNN-LSTM architecture for machine-generated phishing detection in real-time scenarios [38]. [39] has provided further evidence on how large language models are increasingly becoming relevant to cybersecurity by illustrating how transformer-based generative AI systems can enhance phishing email detection using context and semantic reasoning. The hybrid architecture of such models includes the use of CNNs to extract spatial features and Long Short-Term Memory (LSTM) networks to model temporal features. As a result, this allows for detecting patterns that change not only over time but through several interactions. Kulkarni et al. [40] note that the current trend in phishing detection with AI-based systems consists of neglecting the predictive analytics component in favor of classification accuracy. Indeed, [40] states that most studies focus on detecting results of completed phishing attacks, rather than prevention.

2.2 Findings From Systematic Reviews and Surveys

The survey studies offer a broader analysis of phishing detection and social engineering research, concentrating on methodological developments, evaluation strategies, and open problems. Authors in [41] present a detailed overview of phishing website detection techniques and conclude that the majority of recent research utilizes machine learning and deep learning algorithms, which outperform rule-based algorithms due to better performance. In addition, [34] gives an up-to-date survey that illustrates the transition of phishing detection from handcrafted features to transformer architecture. [41] and [34] conclude that machine learning and deep learning algorithms outperform the rule-based systems because of their complexity. Finally, Divakaran and Oest conduct a critical assessment of phishing detection methods and point out that despite the high accuracy of deep learning models, they have poor interpretability and robustness in practical settings [4]. Moreover, Kavva and Sumathi continue the discussion and identify emerging methodologies, including hybrid algorithms and attention-based models, along with existing open issues in scaling and deployment [5].

[3] analyzes phishing website detection methods and observes that although there are multiple methodologies used, most of the techniques employ the same datasets and measurement criteria, thus calling into question the reproducibility of research results. Likewise, Abdillahi et al. have systematically reviewed the phishing classification techniques and concluded that machine learning and deep learning approaches are more accurate than the black-listing and heuristics-based conventional techniques, along with certain issues concerning data sets, feature selection, and model generalization [42]. Asiri et al. note that URL-based and HTML-based phishing detection continue to be predominant areas of study, even though the increasing sophistication of phishing attacks has necessitated more hybrid and adaptive solutions [43]. Furthermore, [44] discusses measures for mitigating phishing attacks and finds that mere technical measures cannot be sufficient without addressing behavior and organizational aspects. [45] advances this theory by considering the phishing literature in terms of electroencephalography (EEG) and eye-gaze tracking techniques, and suggests that both cognitive and behavioral signs could shed light on humans' vulnerability to phishing attempts and also help in creating user-centric defense systems. [21] and [46]

concentrate on the problem of phishing email detection via NLP and deep learning methods, emphasizing that transformer-based architectures perform much better than classical NLP tools because of contextual comprehension abilities. After a synthetic study of 46 papers, six major critical gaps were identified in the research area:

- **Lack of research on data warehousing of phishing and social engineering data:** While it has been shown that data warehousing can be used for analyzing cybercrime and intrusion detection data by [7], there are only a few studies that concern themselves with the application of star-schema-based approaches to phishing and social engineering attack data.
- **Research fragmentation:** Most of the literature covers issues of either the taxonomy of social engineering attacks, phishing detection algorithms, or data warehousing. The limited research concerns the creation of a combined approach integrating structured data storage (star schema), multidimensional analysis (OLAP), and prediction (data mining) for efficient phishing detection.
- **Phishing detection focus:** The research pays more attention to phishing emails than other types of attacks, such as baiting, pretexting, water-holing, vishing, tailgating, and quid pro quo [2].
- **Adversarial robustness problems:** According to Kim et al. [33], phishing schemes constantly evolve with new techniques to avoid detection. Therefore, there is a need to develop detectors that go beyond the static feature-based approach. Yet, adversarial robustness is an underexplored aspect in most works.
- **Little practical validation:** There are very few instances of testing the effectiveness of models in practice; most validations are done using carefully curated datasets [40].
- **Lack of proactive analysis:** [40] note that current studies focus on classification rather than predictive and proactive analysis of threats.

3 Methodology

The paper uses a systematic literature review approach that is guided by the PRISMA 2020 framework. This involves the identification, evaluation, and synthesis of recent scholarly literature regarding phishing attacks, social engineering attacks, cyber security data warehouses, OLAP, and data mining. The process involves four key stages including literature identification, eligibility assessment, screening, and data extraction and synthesis. Relevant articles were obtained from major scholarly databases by use of keywords. The articles were then evaluated through the inclusion and exclusion criteria. The selected articles were coded according to the research questions that include designing a data warehouse using a star schema approach, OLAP approaches to analyze data, and data mining techniques to predict and classify threats. Finally, both quantitative and qualitative synthesis was applied to reveal research trends, advances in methodologies, limitations, and gaps in the cybersecurity data warehouse framework. Figure 1 depicts the detailed systematic paper screening procedure for paper selection.

3.1 Search Strategy and Information Sources

This systematic review conducted a systematic search following the PRISMA 2020 framework. Databases and digital libraries included IEEE Xplore, ACM Digital Library, ScienceDirect, SpringerLink, Wiley Online Library, Scopus, and arXiv. Gray literature sources such as SSRN and Google Scholar were also consulted.

- Search strings combined Boolean operators: (“phishing” OR “social engineering”) AND (“data warehouse” OR “star schema”) AND (“OLAP” OR “multidimensional analysis”) AND (“data mining” OR “knowledge discovery”).
- Time window: 2021–2026
- Language limits: English only

3.2 Eligibility Criteria

The general idea was to incorporate empirical or design-based research that specifically addressed the usage of data warehouse, OLAP, or data mining techniques for phishing and social engineering detection. The research questions (RQ1–RQ3) were in line with the criteria.

3.2.1 Inclusion Criteria

- Published between 2021–2026.
- CORE A/A* conference or SCImago Q1 journal.
- Full-text availability.
- methodology that is empirical or design-based that yields quantifiable results (accuracy, scalability, query performance)
- Direct connection to data warehouse, OLAP, data mining, or phishing/social engineering applications.

3.2.2 Exclusion Criteria

- Outdated Papers(Published before 2021)
- Low-rank venues.
- Publications that are conceptual rather than experimental.
- Non-English publications.
- Papers unrelated to data warehouse analytics or phishing/social engineering.
- No measurable outcomes reported.

3.3 Screening Processes

The publications were screened according to inclusion and exclusion criteria using a PRISMA style flow diagram shown in figure [1](#).

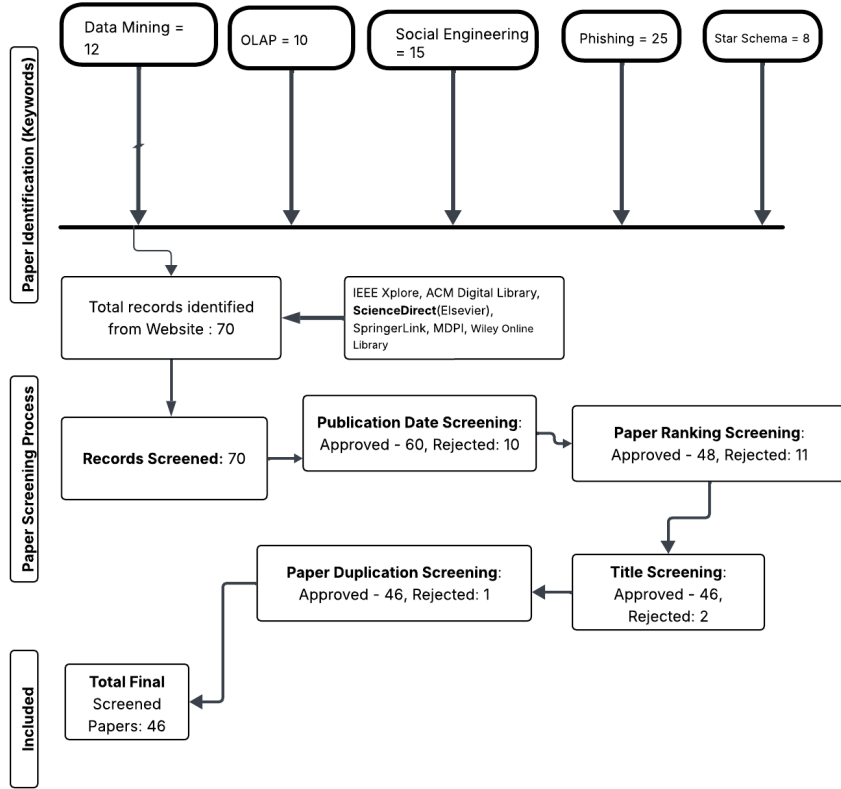


Fig. 1: Systematic Screening Processes For Paper Selection

3.3.1 Publication Date Screening

To guarantee methodological recentness, only articles released between 2021 and 2026 were kept. Total 10 papers were rejected from date screening. The process has been shown in figure 1.

3.3.2 Rank Screening

Venue quality was checked against SCImago (Q1) and CORE (A/A*) rankings. Papers below threshold were excluded. Total 11 papers were rejected from rank screening. The process has been shown in figure 1.

3.3.3 Title Screening

Titles were examined for their applicability to data warehouse/OLAP/data mining and phishing/social engineering. Dual review was used, and consensus was used to break ties. 2 papers were rejected from title screening as shown in figure 1.

3.3.4 Duplication Screening

DOI(Digital Object Identifiers) matching, metadata comparison, and fuzzy title matching were used to find duplicates. The final peer-reviewed journal version was kept when there were several versions. 1 paper was rejected from duplication screening as shown in figure 1.

3.3.5 RQ Match Screening

Studies were mapped to RQ1–RQ3 using a coding schema:

- RQ1: Star schema design for phishing/social engineering data
- RQ2: OLAP techniques for pattern discovery
- RQ3: Data mining methods for prediction/classification

All the papers’ topics have been matched to these three RQs.

3.4 Data Extraction

A standardized extraction form was captured:

- Study metadata (title, year, venue, origin)
- Dataset details (phishing URLs, email corpora, social engineering case studies)
- Methods (data warehouse design, OLAP operations, ML/data mining algorithms)
- Baselines (traditional detection methods)
- Metrics (accuracy, precision, recall, query response time)
- Limitations (dataset bias, scalability issues)

Synthesis plan:

- Quantitative synthesis: Meta-analysis of detection accuracy and OLAP query performance.
- Qualitative synthesis: Thematic analysis of star schema designs and integration strategies.

4 Results

The following section discusses the results derived from the systematic review in relation to the three research questions covering publication trends, methodological approaches, and technology usage across the review papers and portrays into the graphs. First, an overview of the paper selection process and the features of the selected 46 papers is provided, along with the examination of publication trends and the relevance of the research questions. Afterward, it synthesizes the results of the studies on the use of star schema-based data warehousing, OLAP technology, and data mining methods for phishing and social engineering detection with respect to the key trends and limitations found from the selected papers in the following subsections.

4.1 Paper Elimination Summary

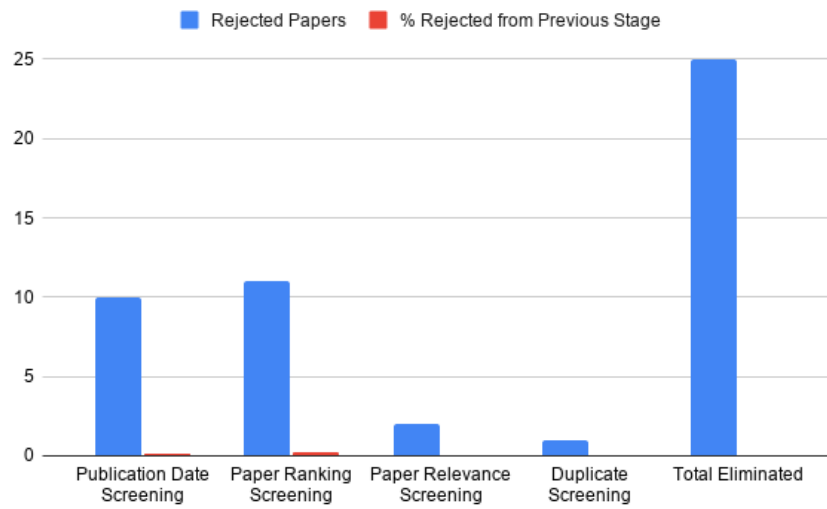


Fig. 2: Paper Elimination Process Breakdown.

The selection of papers for the study has been shown through the following Figure 2. A total of 70 documents was collected through specific keywords in different academic sources. After this, the papers were arranged on the basis of duplication, relevance, ranking, and date of publication. At last, 46 quality papers were selected for literature review after eliminating insufficient papers at each step.

4.2 Demographic Analysis of the Selected Research Papers

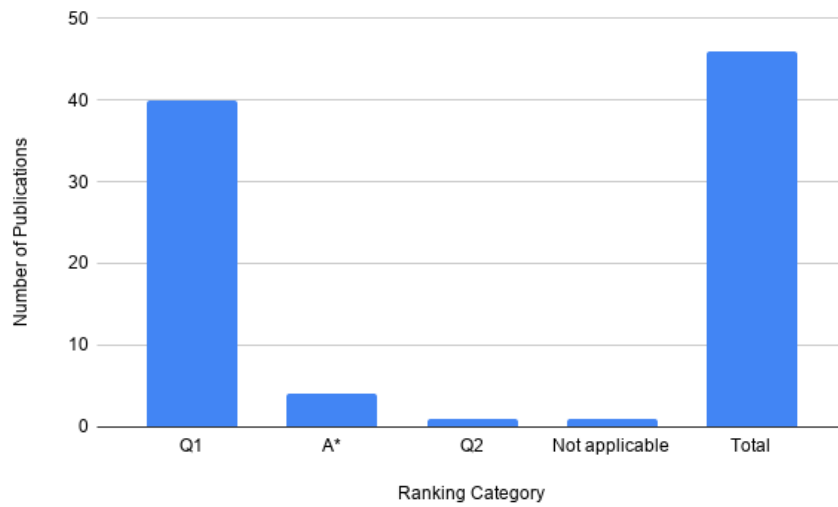


Fig. 3: Demographic Analysis of the Selected Research Papers.

As shown in Figure 3, the chart represents the ranking pattern of the selected research articles. Four research papers have been published in A* ranking journal; however, nearly all the research (40 papers) has been published in Q1-ranked journals. It can be said that the reliability of the study results was enhanced due to the fact that the literature review was based on credible research sources.

4.2.1 Publication Trends Over the Years

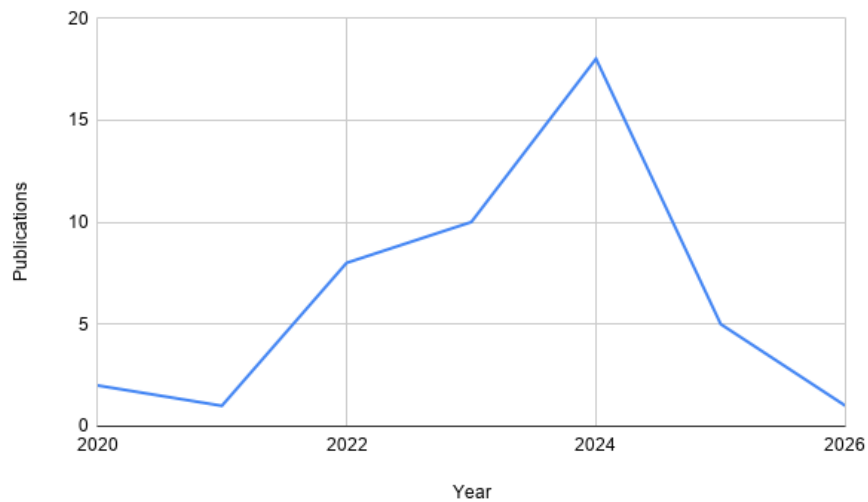


Fig. 4: Publication Trends Over the Years.

The publication pattern from 2020 to 2026 is provided in this chart as illustrated in Figure 4. It is clearly seen that the number of articles has been increasing consistently, with the peak point being 2024. Despite the slight decrease after 2024, the overall trend remains increasing. This means that during the mid-year period there was a massive growth of the research activity.

4.2.2 Distribution of Papers by Database

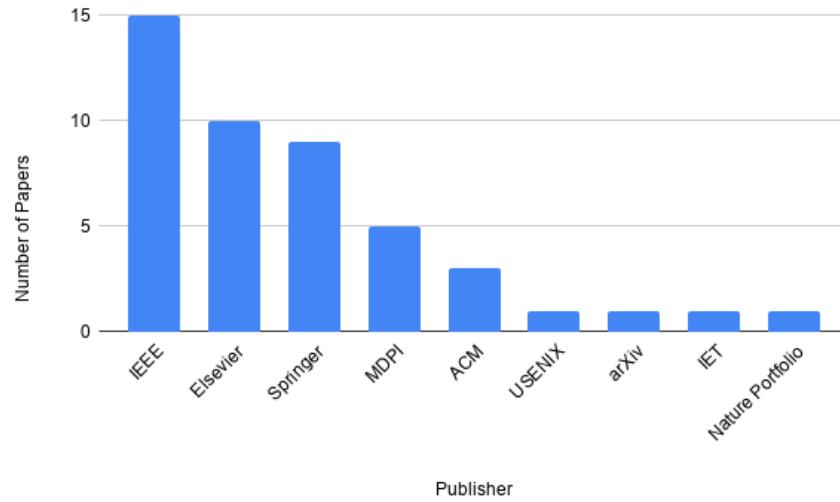


Fig. 5: Distribution of Papers by Database.

As seen from Figure 5, the chosen scholarly papers have been collected from different academic databases applied in this work. The largest number of published materials has come from the database IEEE Xplore, followed by ScienceDirect and SpringerLink. ACM Digital Library has provided the necessary number of scholarly sources too. As can be seen, the chosen papers have been collected mostly from reputable academic databases and digital libraries.

4.2.3 Geographical Distribution of Research Publications

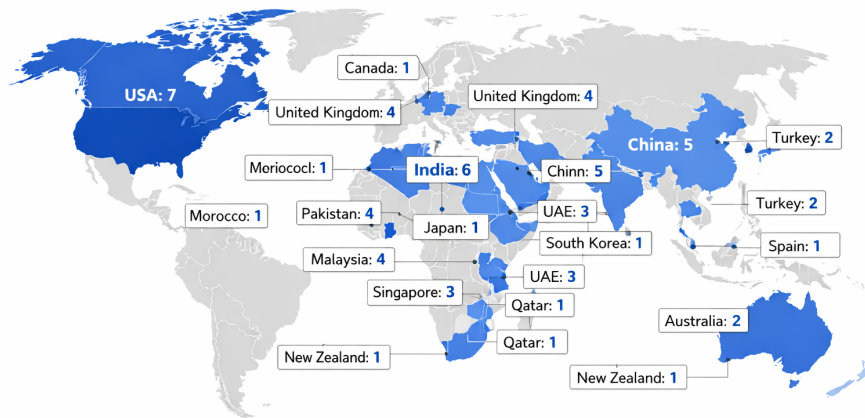


Fig. 6: Geographical Distribution of Research Publications.

From figure 6 above, the graph indicates how widely research papers are published geographically based on the countries of the researchers. The USA leads in the production of research papers with 7 papers, followed by Saudi Arabia and India with 6 papers. There is a significant amount of research papers published from China and Pakistan. All in all, it can be seen that although there is dispersion of research across many countries, it is mainly concentrated in certain parts only.

4.2.4 Research Publications by Focus Area

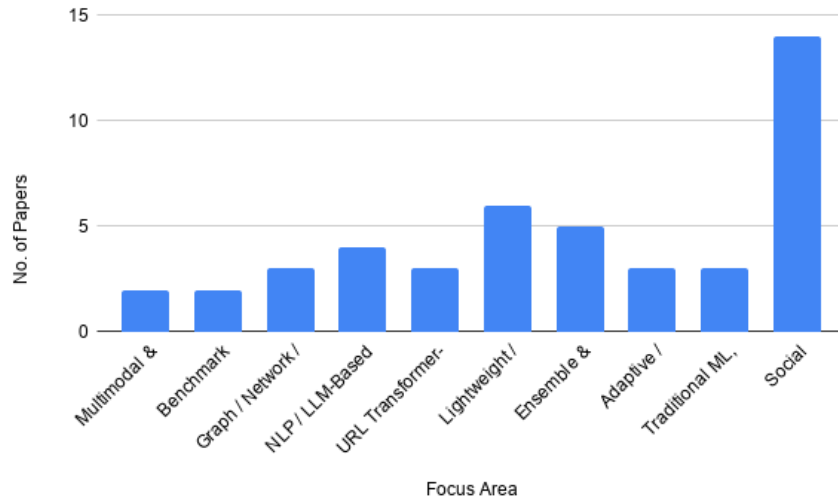


Fig. 7: Research Publications by Focus Area.

As illustrated in Figure 7, the chart shows the distribution of 46 research papers across different phishing and cybersecurity focused areas. It helps to identify which topics are most commonly studied, such as social engineering, machine learning models, and hybrid detection systems. The data also highlights emerging areas like multimodal and LLM-based phishing detection, which are gaining attention. Overall, it gives a clear overview of current research trends and priorities in the field.

4.3 Methodological Approaches: Research Strategies and Techniques

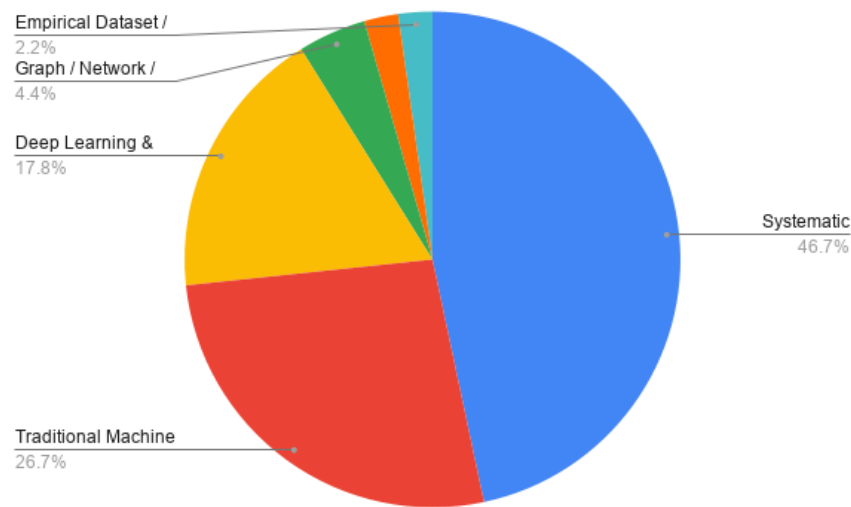


Fig. 8: Distribution of Research Methodologies Used

As can be seen in Fig. 8, this graph represents the distribution of the 46 papers based on different study techniques used, such as deep learning techniques, machine learning algorithms, and systematic literature reviews. Having an idea of which study techniques are used widely in phishing research is important. From the graph, one can see that the field is mainly concentrated on review studies and machine learning techniques, but some other advanced techniques are rare but emerging.

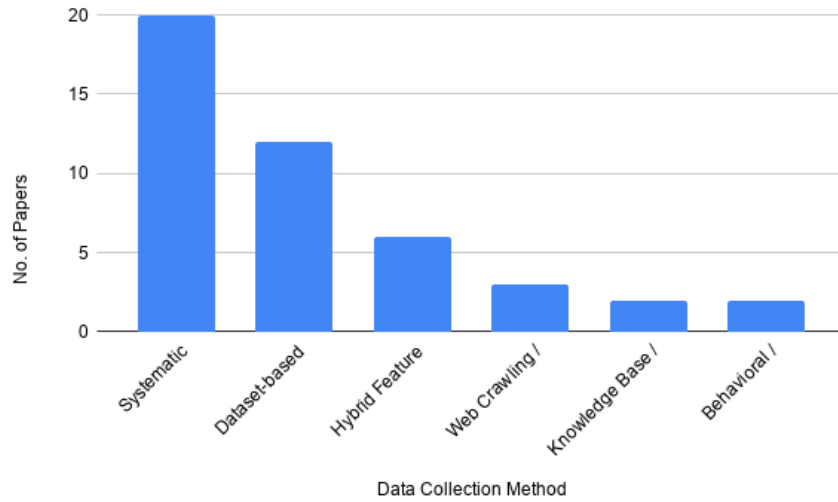


Fig. 9: Data Collection Methods.

As it can be observed from Figure 9, this graph shows the number of ways through which researchers employ to collect data in the case of 45 articles, such as web crawlers, behavioral experiments, real world data, and systematic reviews. It should be noted that although there are fewer researches that conduct real time data collection or data related to human behavior, the majority of them are related to literature reviews and existing data sets.

4.4 Cutting-Edge Technologies

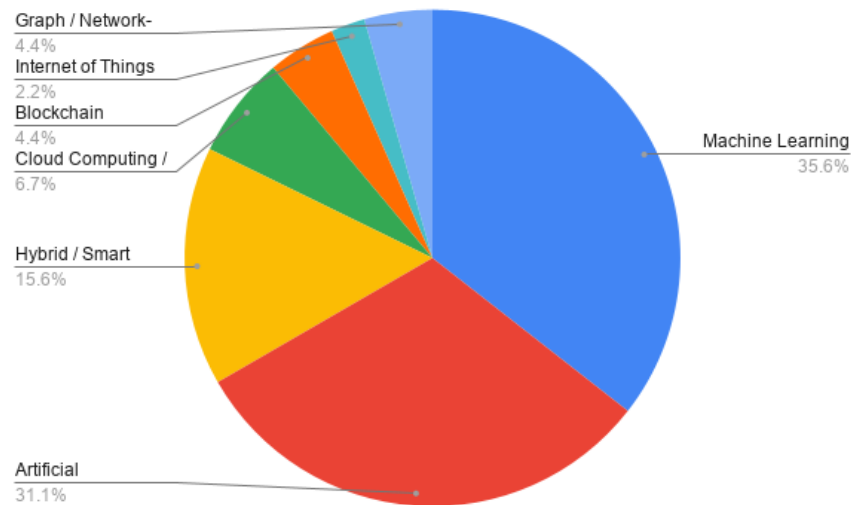


Fig. 10: Technologies Used In Research.

As shown in Figure 10, this diagram shows the various techniques used in 46 research papers on cybersecurity and phishing. It highlights that while comparing with other technologies, the use of machine learning and artificial intelligence is the most common technique. Furthermore, it shows the increasing trend of advanced technologies such as blockchain, IoT, and cloud-based technologies.

4.5 Key Research Themes: Key Outcomes from Selected Studies

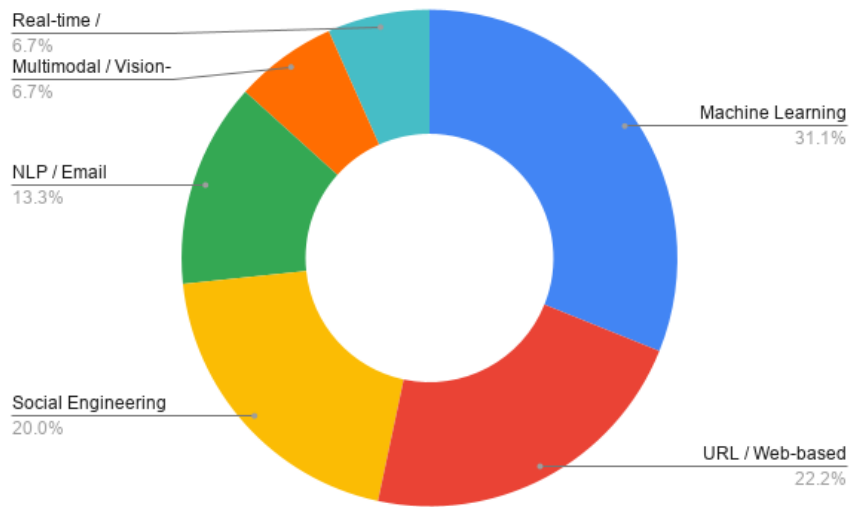


Fig. 11: Breakthrough Findings: Key Outcomes from Selected Studies.

As illustrated in Fig 11, this figure depicts the main research areas covered in the 46 selected publications on cybersecurity and phishing detections. It clearly illustrates that whereas there is less but growing research interest in the study of other domains such as the social engineering domain, NLP techniques, and multi-faceted approaches, most of the research interests lie in ML, DL, and URL-based methods.

4.6 Barriers to Implementation: Challenges and Limitations

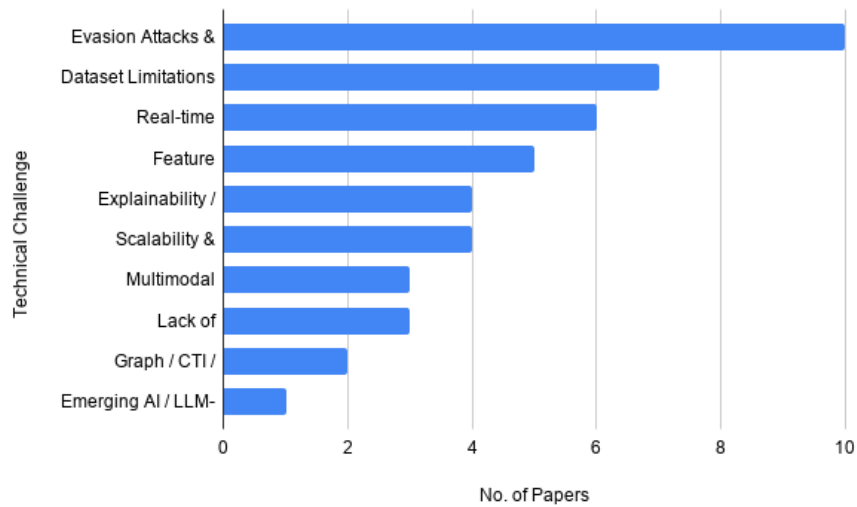


Fig. 12: Barriers to Implementation: Challenges and Limitations.

As depicted in Figure 12, this graph shows the key technical challenges in the 46 articles studied in relation to cybersecurity and phishing detection. The challenges that have been stressed as the most crucial in this area are that of evasion attacks, zero-day phishing, and constraints on the data sets used to train the models.

4.7 Future Prospects: Research Gaps and Emerging Directions

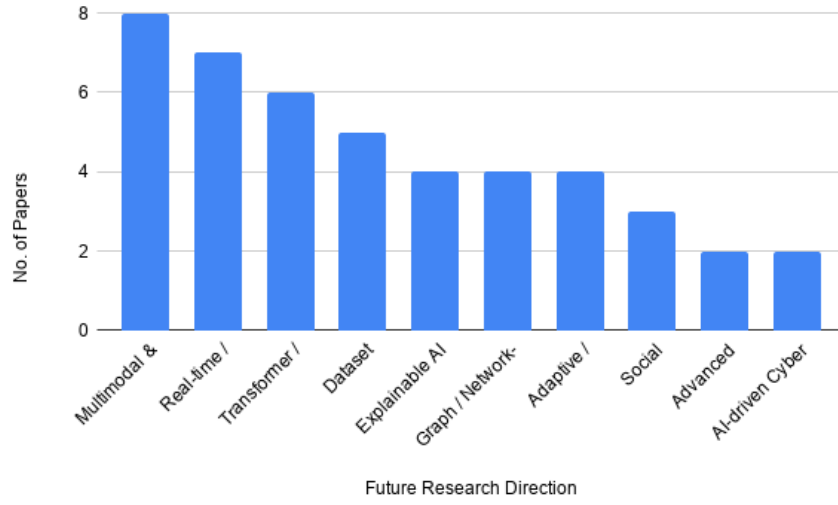


Fig. 13: Future Prospects: Research Gaps and Emerging Directions.

Figure 13 depicts potential avenues of future research that can be deduced from 46 studies on phishing detection research. This includes a heavy focus on transformer and LLM-based approaches, lightweight and real-time models, as well as multi-modal models. Other emerging areas include better data sets and adaptive learning techniques.

4.8 Impact on Society: Practical and Social Implications

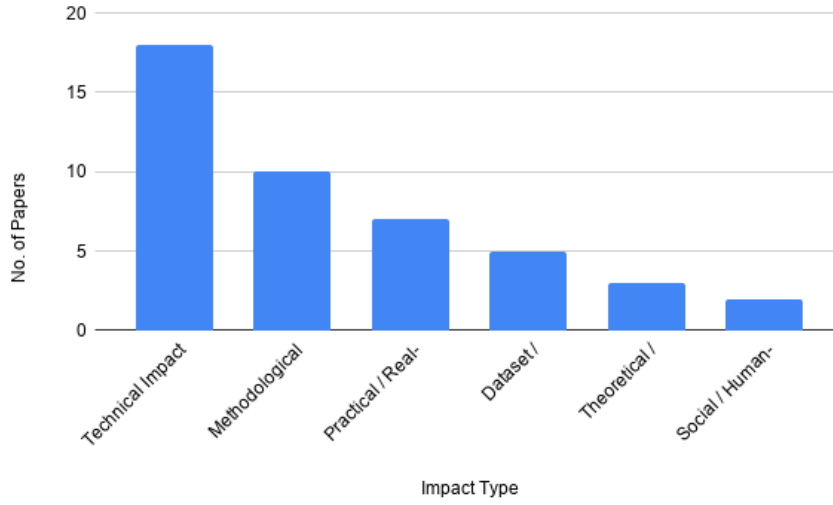


Fig. 14: Impact on Society: Practical and Social Implications.

As illustrated in Figure 14, the graph presents a categorization of research papers by the nature of the impact they have made on cybersecurity and phishing research. It clearly demonstrates that most of the papers make an impact through methodology review in the form of surveys and taxonomies after technological innovations such as model development and algorithms. Moreover, it illustrates small but important impacts made on data sets, theoretical development, people security research, and practical application.

4.9 Contribution and Research Questions Matching

In this section, it is examined how the contributions presented in the above mentioned studies are related to the research questions formulated above. The contributions are sorted according to areas like detection techniques, datasets used, evaluation methods, analysis methods, etc., and how each research question is addressed by the chosen papers.

Coding Scheme:

Contribution categories were systemically defined and assigned to the corresponding RQs as follows:

- Novel algorithm → New detection/classification models (CNN, transformer, hybrid ML/DL).

- Dataset/benchmark → Creation or analysis of phishing/social engineering datasets (VirusTotal, ccTLD scale).
- Evaluation framework → Systematic reviews, surveys, or comparative studies.
- Application/deployment → Real-time systems, IoT-based monitoring, or large-scale deployment.
- Interpretability/calibration → Explainable AI, security patterns, psychology-based defenses.

Scoring rules:

- Fully answered → Directly addresses the RQ with validated methods or comprehensive survey.
- Partially answered → Provides insights but lacks full validation, scalability, or integration.
- Not answered → Mentions topic but no substantive contribution.

4.9.1 Overview of Contributions

One feature of the present review that distinguishes it from other reviews in the field is its narrow focus, which combines five themes related to phishing and social engineering: attack taxonomies, phishing detection methods, cyber threat intelligence, data warehousing techniques, and AI-based predictions. The present review draws on methodological developments in machine learning, deep learning, transformers, and data warehousing. Through offering a synthesized perspective on how data management, analytics, and prediction can be used for cybersecurity knowledge discovery, the present review highlights the methodological weaknesses of existing research in the field, demonstrates the fragmentation of current studies, and presents the opportunities for future research on the integration of social engineering taxonomies, phishing detection methods, and threat intelligence frameworks. Here, the main contributions of the paper are discussed as follows:

- **Methodological advances:** Most papers propose ML/DL models (CNN, LSTM, transformers, hybrid ensembles). Metrics: accuracy, F1-score, precision/recall. Representative: CNN-MHSA, URLNet, KnowPhish. Maturity: exploratory to validated.
- **Datasets/benchmarks:** Large-scale studies (VirusTotal reports, ccTLD scale) provide empirical grounding. Maturity: validated, widely cited.
- **Evaluation frameworks:** Systematic literature reviews and surveys (Phishing Classification Techniques, Social Engineering Attacks Prevention) synthesize evidence. Maturity: validated but incremental.
- **Application/deployment:** Real-time detection systems (PhishingRTDS, hybrid CNN-LSTM frameworks) and IoT warehouse schema proposals (RQ1). Maturity: exploratory prototypes.
- **Interpretability/calibration:** Security pattern frameworks, explainable transformer models, psychology-based defenses. Maturity: emerging, exploratory.

Appraisal: Novel algorithms dominate, but deployment and interpretability remain underdeveloped. Dataset contributions strongly support RQ1, while OLAP/knowledge discovery (RQ2–RQ3) are partially addressed.

4.9.2 Research Question Matching Analysis

Table 1 presents the analysis of the matching of research questions, and the mapping of the study’s research questions to the questions of other studies is shown in Figure 15.

Table 1: Analysis of Research Questions Matching

Research Question	Fully Answered	Partially Answered	Not Answered
RQ1: Star schema warehouse for phishing/social engineering	12	18	5
RQ2: OLAP techniques for pattern discovery	9	22	4
RQ3: Data mining for threat prediction/classification	27	14	3

5 Comparison and Analysis

This section investigates the comparative analysis between the results obtained from this systematic review and the results of surveys and reviews already available. The comparisons are regarding the extent and methodology used in the previous studies especially focusing on the coverage of detection methods for phishing, social engineering, data warehousing, OLAP and data mining approaches. It highlights that current studies have mainly focused on the detection and classification of phishing attacks, whereas the inclusion of structured data warehousing and multidimensional analysis techniques is relatively rare. Consequently, this paper justifies the necessity of developing a unified framework based on a star-schema approach for cybersecurity knowledge discovery. Table 2 indicates the comparison and analysis of this study and other systematic surveys.

6 Discussion

This section discusses the major findings, implications, challenges, and limitations across review papers.

6.1 Key Findings and Implications

In this systematic literature review, recent studies carried out from 2021 to 2026 on phishing detection, social engineering attack techniques, cyber threat intelligence, data warehousing, OLAP techniques, and data mining approaches have been analyzed. It can be concluded that machine learning and deep learning approaches reign supreme in today’s phishing detection due to the high detection accuracies of transformer-based architecture, CNN, group learning approach, and hybrid architecture. The necessity of coping with the increasing complexity of online security threats is evident in the recent studies where explainable AI, large language models, and modal phishing detection techniques have been explored.

Table 2: Comparison of Existing Reviews and the Present Study

Reference	Domain / Focus	Key Findings and Techniques	Identified Gaps
Safi et al. [41]	Phishing Website Detection (SLR)	Reviews AI, ML, and DL techniques for phishing website detection, highlighting feature-based and learning-based approaches.	Limited focus on multimodal detection, unified benchmark datasets, and explainable models.
Salloum et al. [42]	Phishing Email Detection and NLP	Analyzes NLP, ML, and DL methods for phishing email detection, with emphasis on text-based feature extraction.	Limited handling of multilingual and evolving phishing content; explainability challenges remain.
Syafitri et al. [10]	Social Engineering Attack Prevention	Examines prevention strategies including awareness, training, and technical countermeasures.	Human factors remain underexplored, with limited integration of AI-driven adaptive prevention systems.
Thomopoulos et al. [45]	EEG and Gaze-Based Phishing Detection	Investigates behavioral and physiological indicators, including EEG and eye-tracking, for phishing detection.	Small-scale studies, limited real-world deployment, and high implementation complexity.
Naqvi et al. [44]	Phishing Attack Mitigation (SLR)	Reviews mitigation strategies, user awareness, technical defenses, and organizational practices.	Limited evaluation of emerging AI-based mitigation methods and long-term effectiveness.
This Work (2026)	Unified AI-Powered Phishing Detection Framework	Integrates URL, webpage text, OCR/image analysis, transformer embeddings, and XAI for phishing detection.	Bridges existing silos through multimodal detection, explainability, and standardized evaluation.

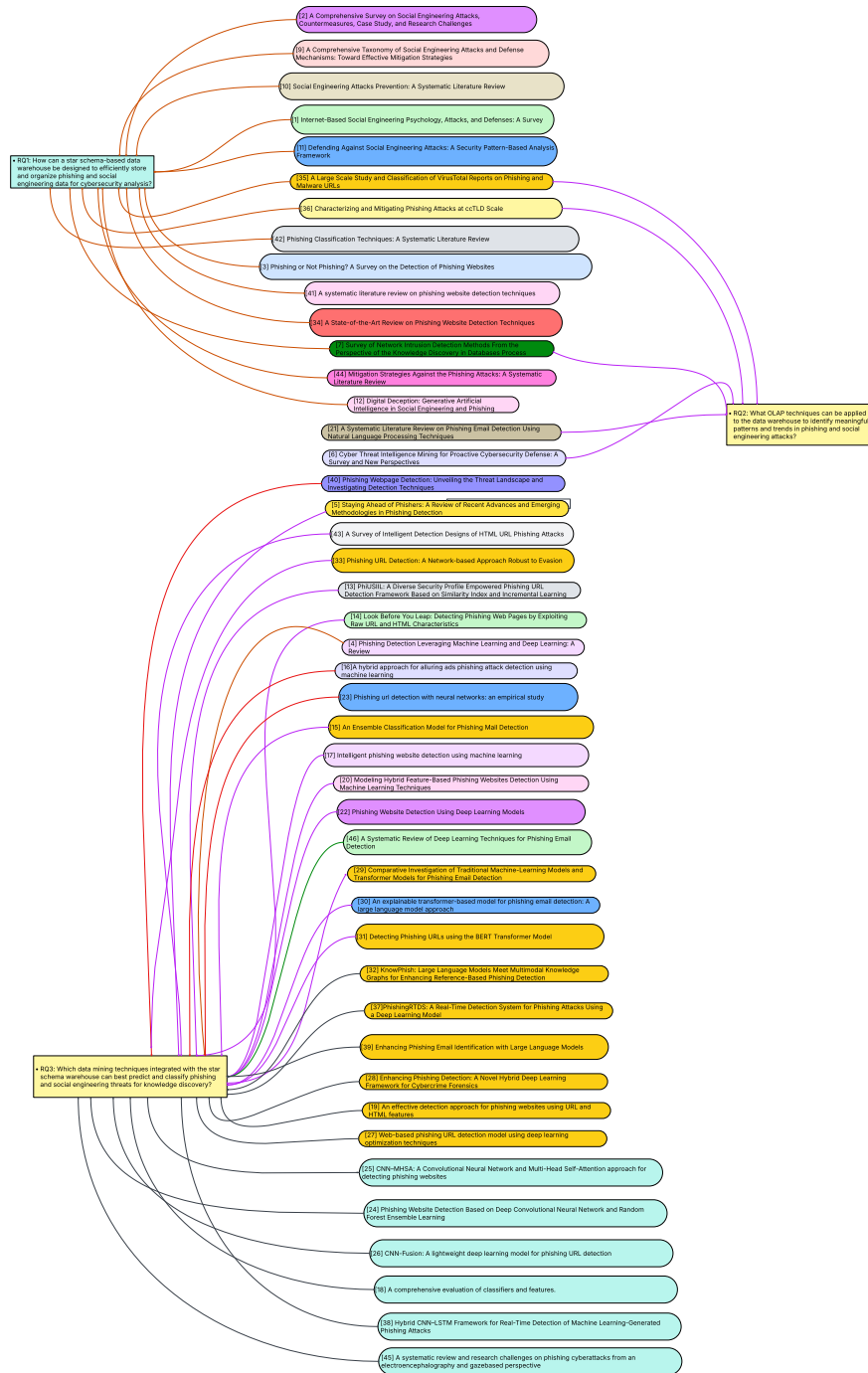


Figure 15: Systematic Screening Process for Paper Selection

Furthermore, the paper suggests that despite much development in the area of classification and detection of phishing, there has been very little work done with regard to storing data, creating a star schema, performing Online Analytical Processing (OLAP), and performing predictive data mining. In other words, the studies conducted were not concerned about the development of adaptive threat intelligence and knowledge creation in the long run, but rather concentrated on post-phishing attack detection. From a practical perspective, the findings imply that enterprises could enhance their cybersecurity protection through the use of advanced detection technologies powered by AI as well as incorporating cyber threat intelligence within their operational decision-making processes. Threat monitoring, pattern recognition, strategic planning, and preemptive defense from emerging phishing and social engineering attacks could all be facilitated using the proposed approach to data warehousing and analysis.

6.2 Challenges and Limitations

However, there are still several gaps in spite of some positive trends have been observed in the literature. One of the most significant problems is the division of the research on the problem under discussion into different areas. Most research does not deal with the integration of a phishing detection algorithm, attack classification system, or cybersecurity analysis into a coherent system – most research focus only on these aspects. Thus, there are no many coherent systems, which could aid in both knowledge discovery and detection. The other barrier is the reliance on specifically chosen datasets in the evaluation process of models. While most proposed methods perform exceptionally well within experimental settings, there is inadequate evidence available regarding their effectiveness during practical application. The constantly changing nature of the phishing attacks, with techniques designed to evade detection mechanisms, is likely to make the practical implementation even harder in the future and reduce the generalizability of the idea. The restrictions in this review are numerous too. Firstly, only English papers published in years from 2021 to 2026 have been considered in the review process, possibly omitting relevant works from other times or languages. Secondly, only reputable sources, i.e. CORE A/A* conference publications and SCImago Q1 journal publications, have been considered; consequently, many findings from creative but less-known projects have been omitted. Lastly, the scope of comparative research in regards to Research Questions 1 and 2 was limited due to few relevant works on data warehouse and OLAP in the field of phishing and social engineering.

7 Conclusion and Future Directions

The latest advancements in the field of phishing detection, social engineering protection, cyber threat intelligence, data warehousing, OLAP, and data mining have been analyzed thorough the study. The findings reveal that phishing detection technology has been vastly enhanced by machine learning, deep learning, converter models, and the combined application of various AI tools. Nevertheless, the blend of structured data

management, multi-dimensional analytics, and prediction through knowledge discovery remains a somewhat neglected topic in comparison to the problem of classification and detection.

The opportunities for further research have been identified in the course of the evaluation process, in particular, in developing a common framework for applying advanced AI-based prediction models, data warehousing techniques, cyber threat intelligence, classification of phishing and social engineering, and OLAP analysis. However, the main results achieved through this study emphasize the importance of having a reliable, scalable, understandable, and feasible system for dealing with rapidly evolving scenarios. Future research efforts need to be directed towards the development of a multi-purpose approach, adversarial robustness, implementation into reality, and finally the creation of a complete cybersecurity analytics system.

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